Design Document for Hostel Management and Complaint Registration System

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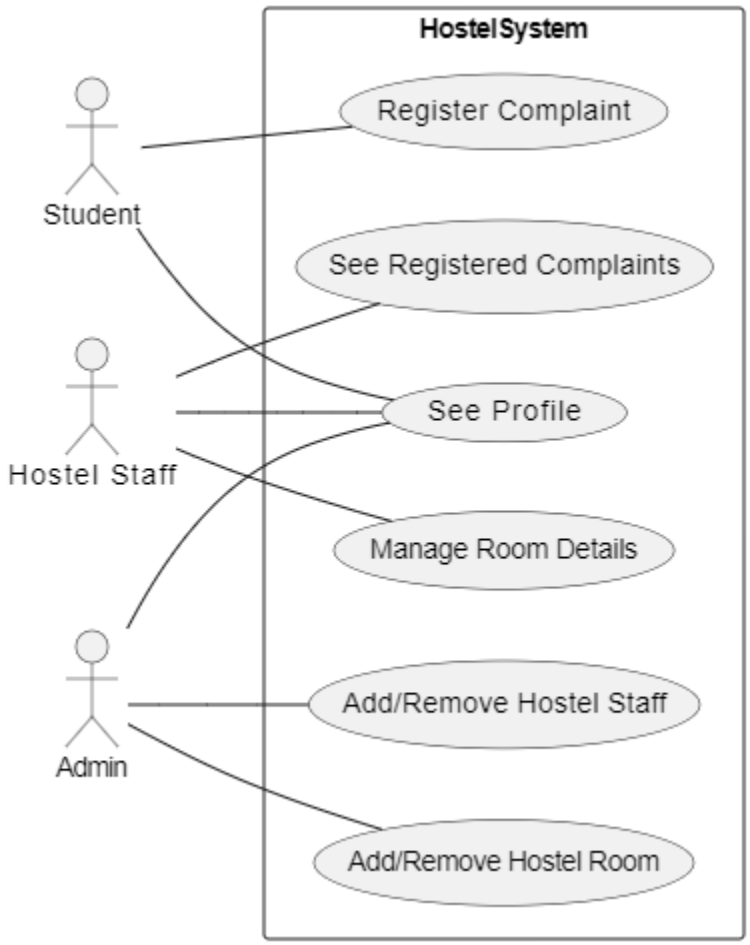
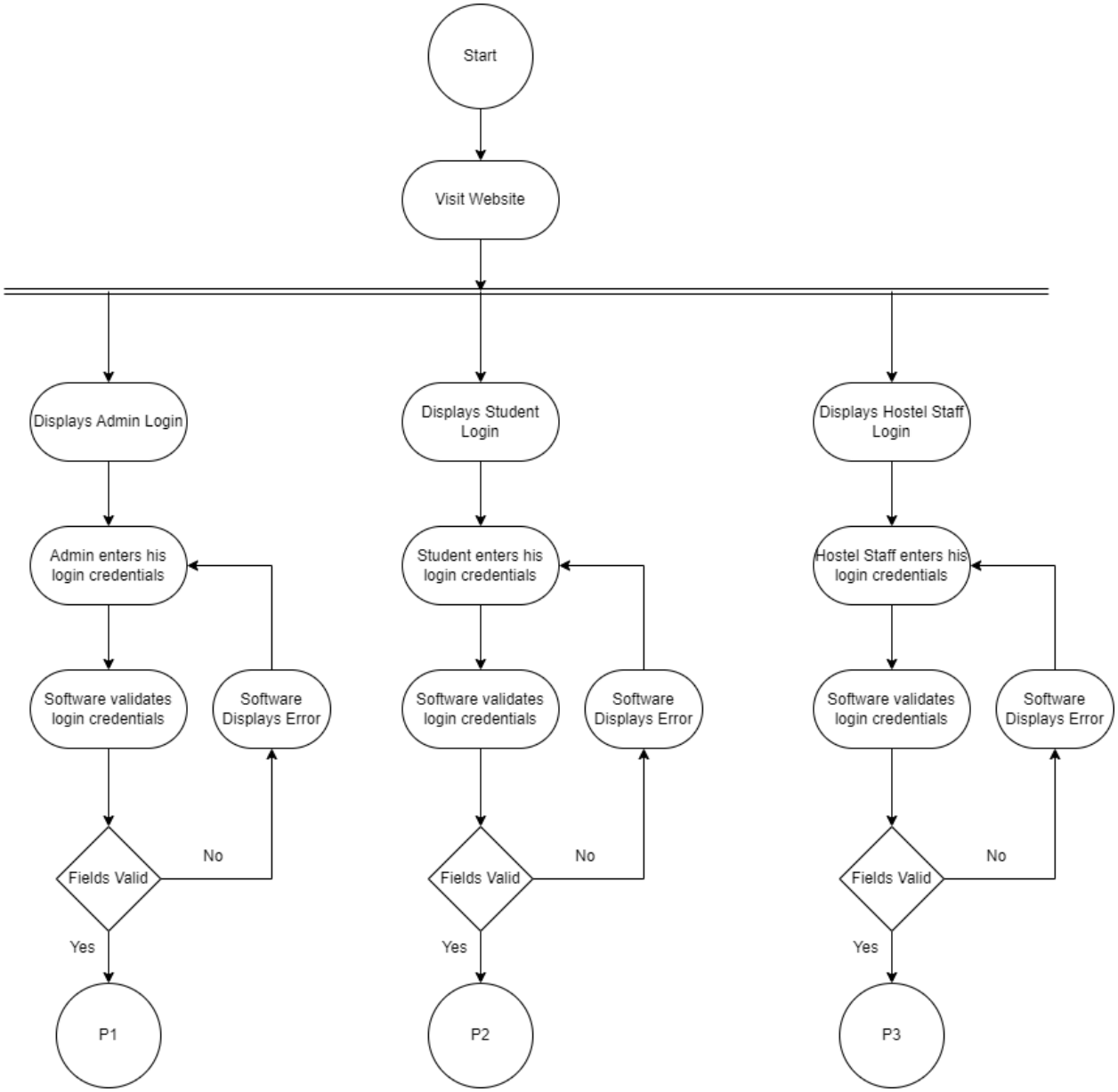
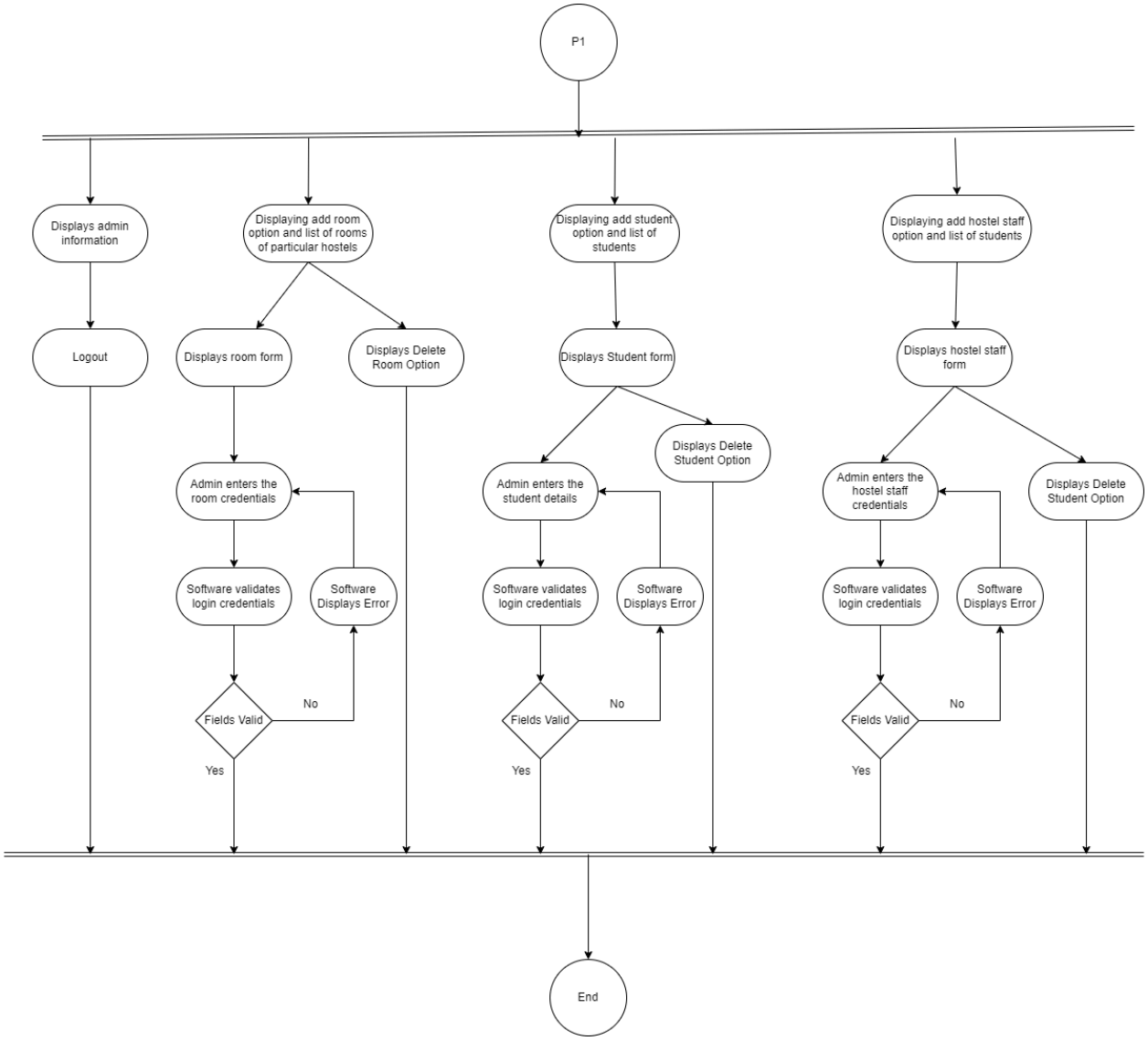
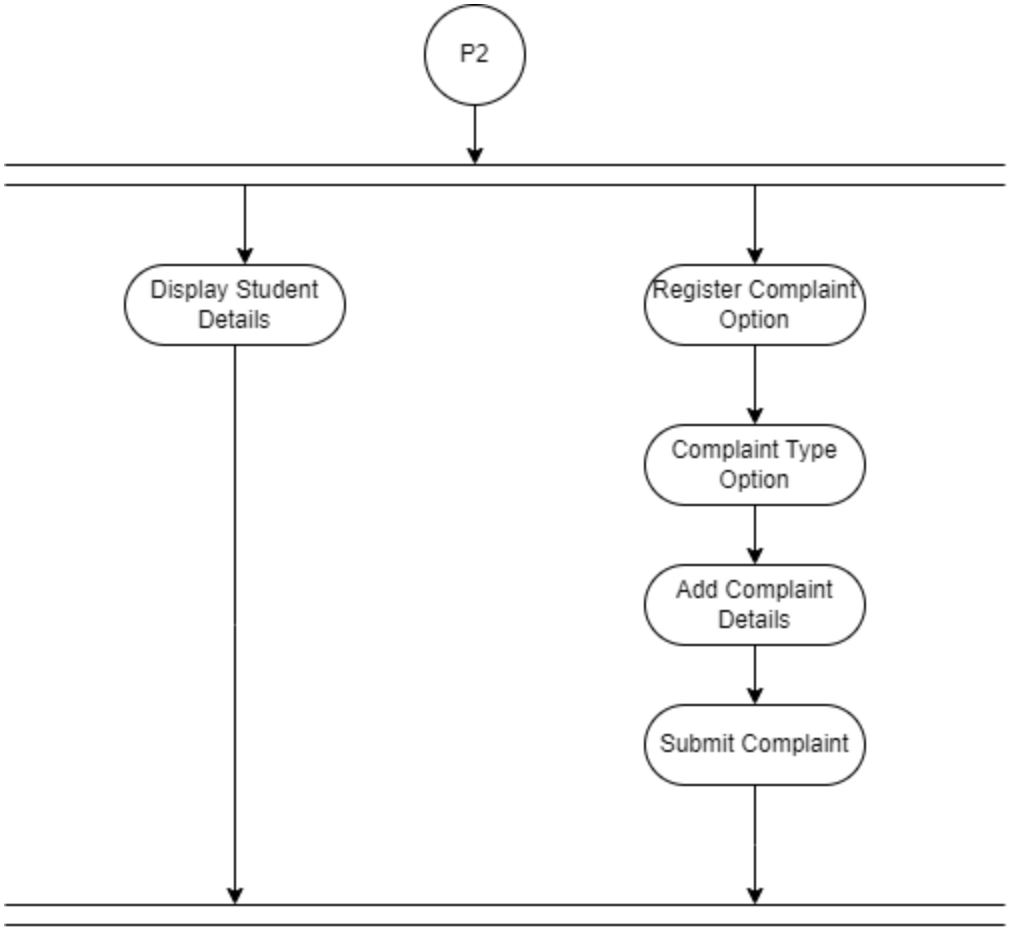
Use Case Diagram:

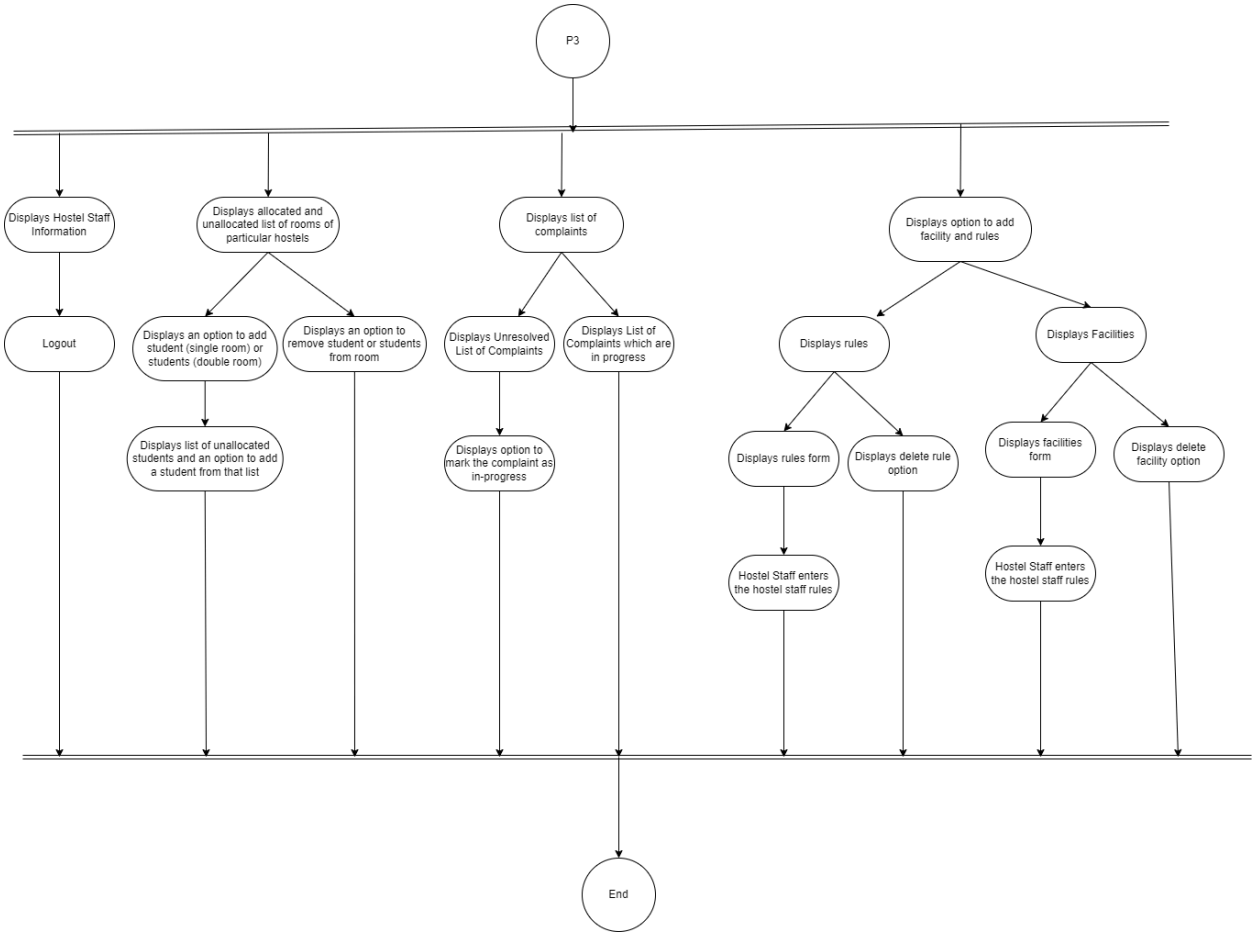


Elaborating Use Case:

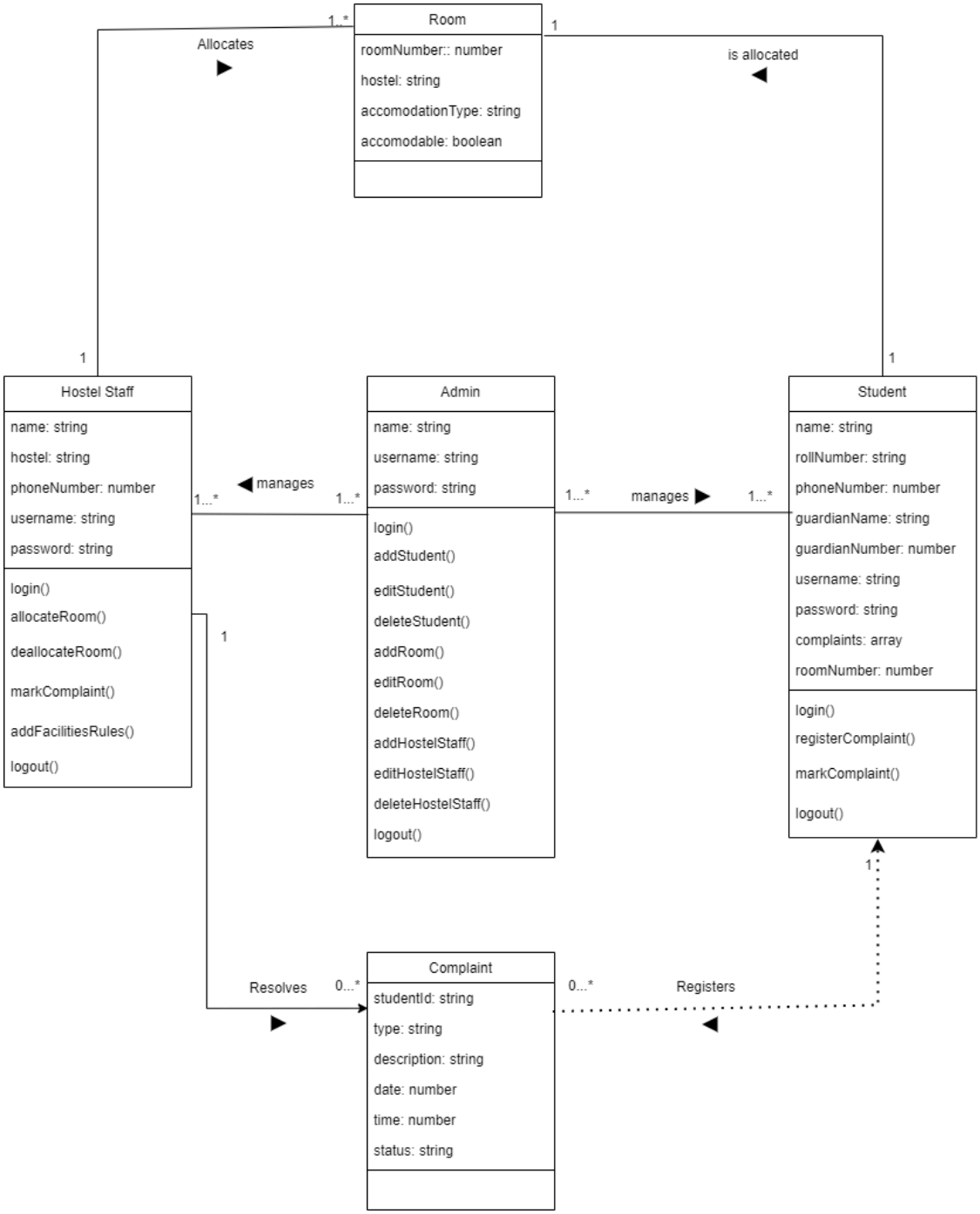



P1
Displays admin information
Displaying add room option and list of rooms of particular hostels
Displaying add student option and list of students
Displaying add hostel staff option and list of students
Logout
Displays room form
Displays Delete Room Option
Displays Student form
Displays hostel staff form
Displays Delete Student Option
Admin enters the room credentials
Admin enters the student details
Admin enters the hostel staff credentials
Displays Delete Student Option
Software validates login credentials
Software Displays Error
Software validates login credentials
Software Displays Error
Software validates login credentials
Software Displays Error
Fields Valid
No
Yes
Fields Valid
No
Yes
Fields Valid
No
Yes
End

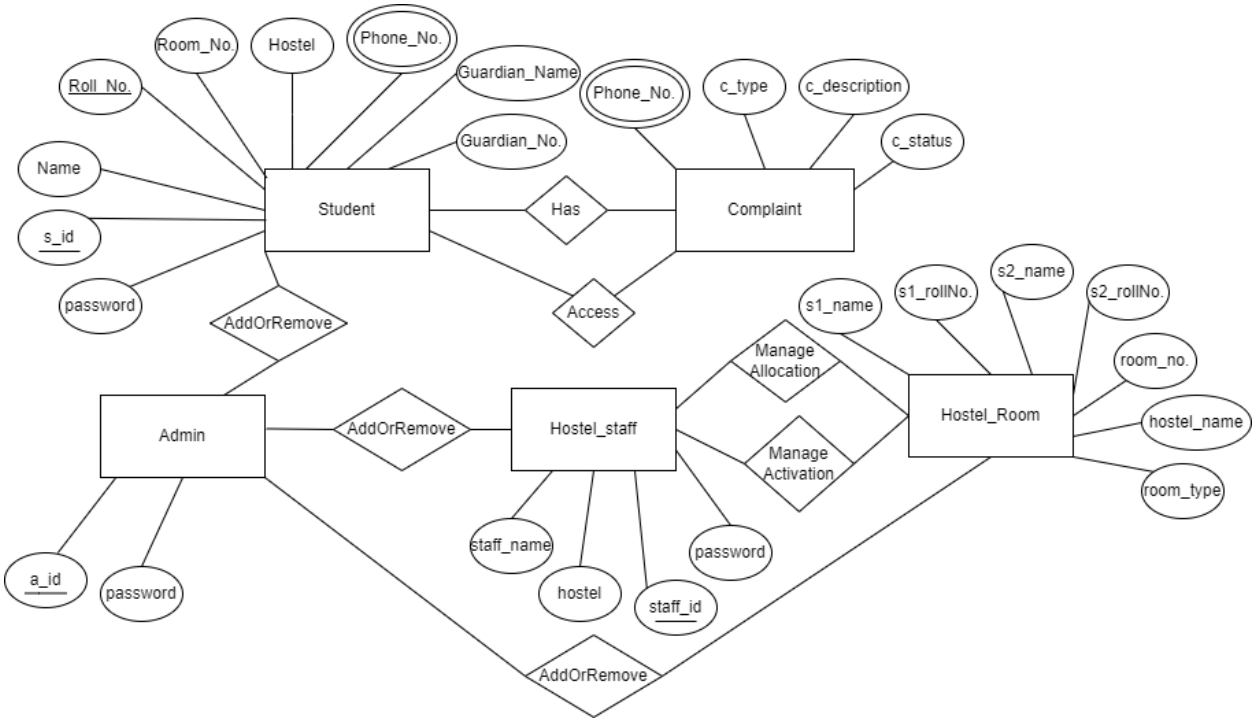

P2
Display Student Details
Register Complaint Option
Complaint Type Option
Add Complaint Details
Submit Complaint



Class Diagram:

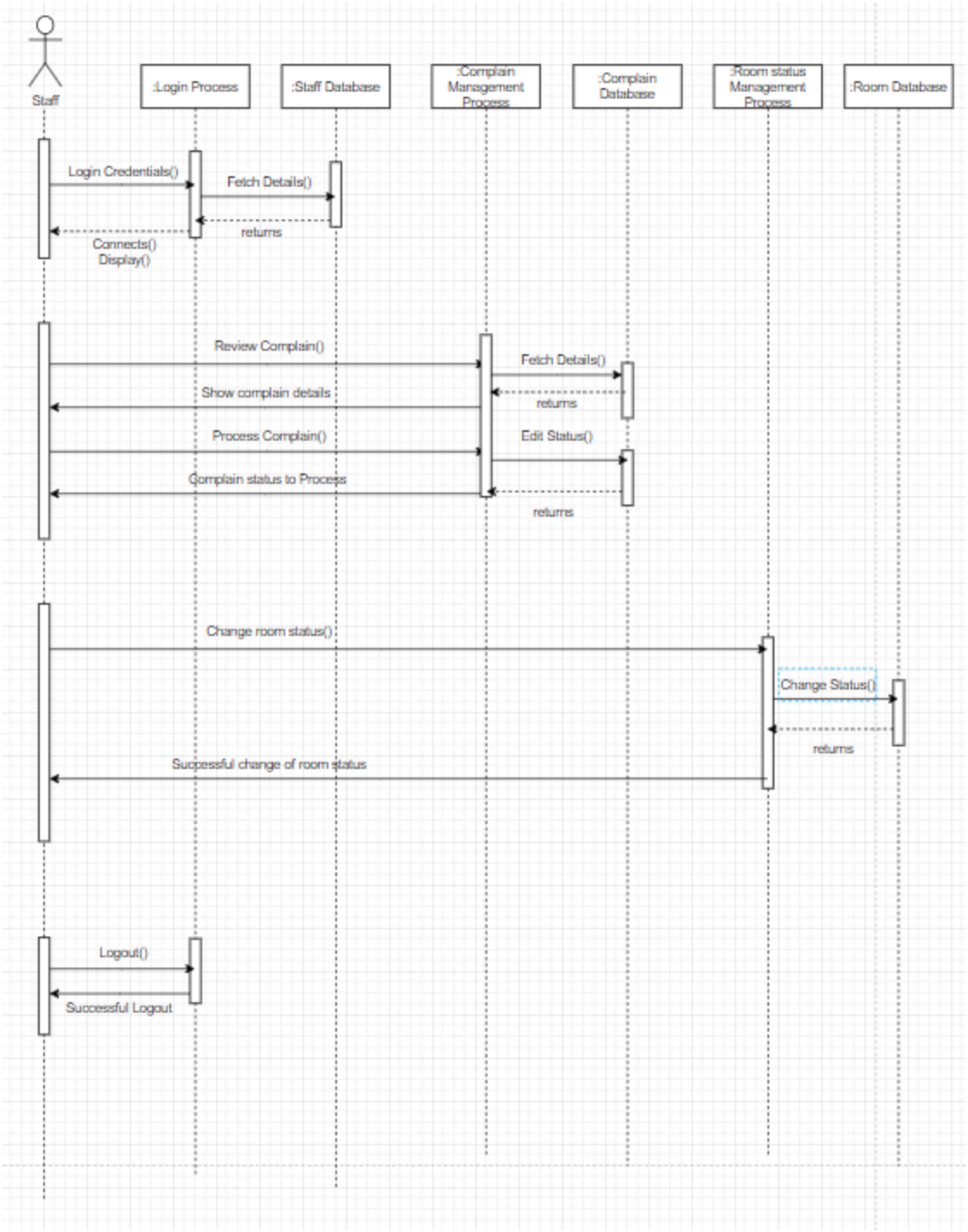

Room
roomNumber:: number
hostel: string
accomodationType: string
accomodable: boolean
Allocates
1...*
1
is allocated
1
1
Hostel Staff
name: string
hostel: string
phoneNumber: number
username: string
password: string
login()
allocateRoom()
deallocateRoom()
markComplaint()
addFacilitiesRules()
logout()
manages
1...*
1...*
Admin
name: string
username: string
password: string
login()
addStudent()
editStudent()
deleteStudent()
addRoom()
editRoom()
deleteRoom()
addHostelStaff()
editHostelStaff()
deleteHostelStaff()
logout()
1...*
manages
1...*
Student
name: string
rollNumber: string
phoneNumber: number
guardianName: string
guardianNumber: number
username: string
password: string
complaints: array
roomNumber: number
login()
registerComplaint()
markComplaint()
logout()
1
1
Resolves
0...*
Complaint
studentId: string
type: string
description: string
date: number
time: number
status: string
0...*
Registers

ER Diagram:

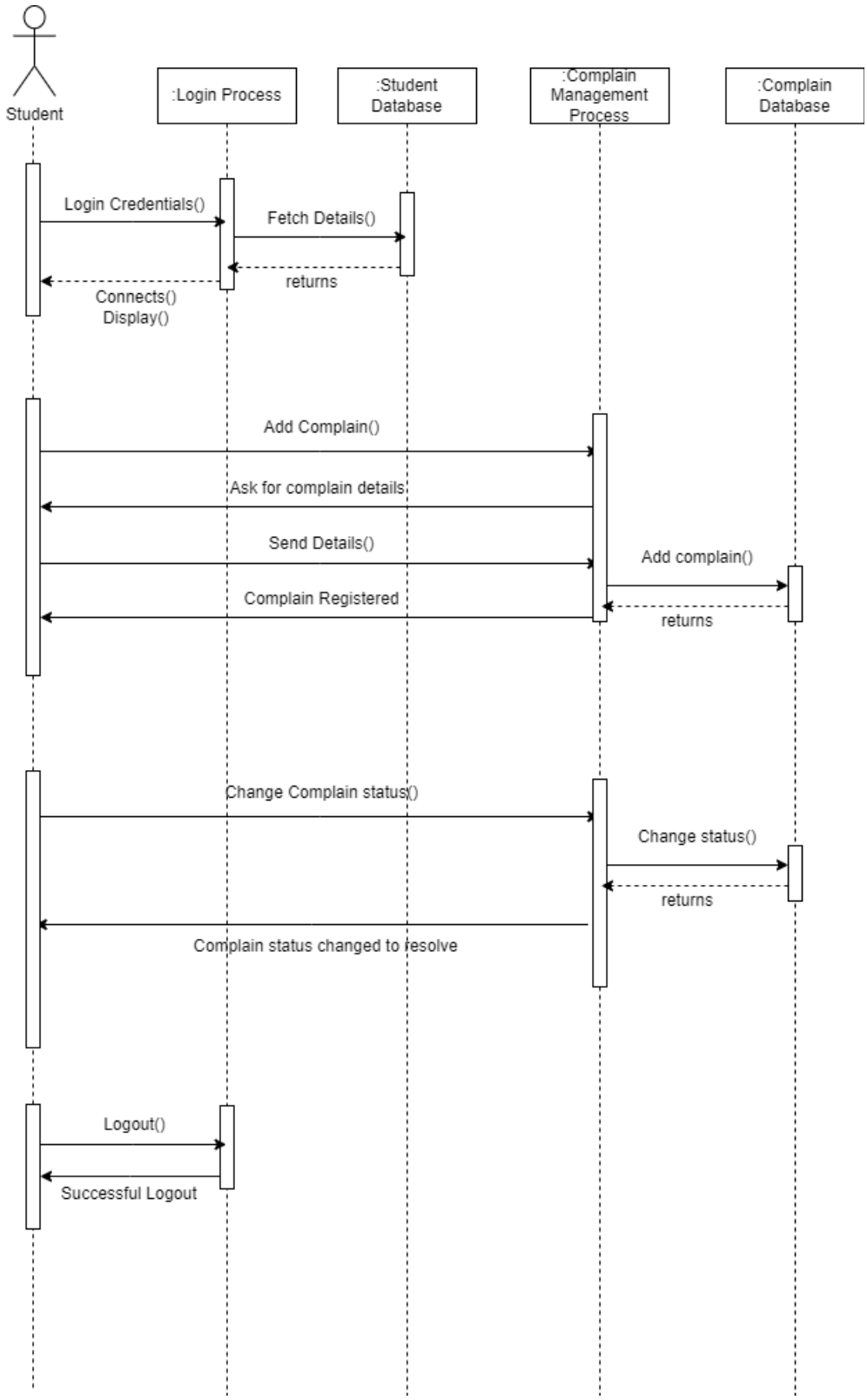


Sequence Diagrams:

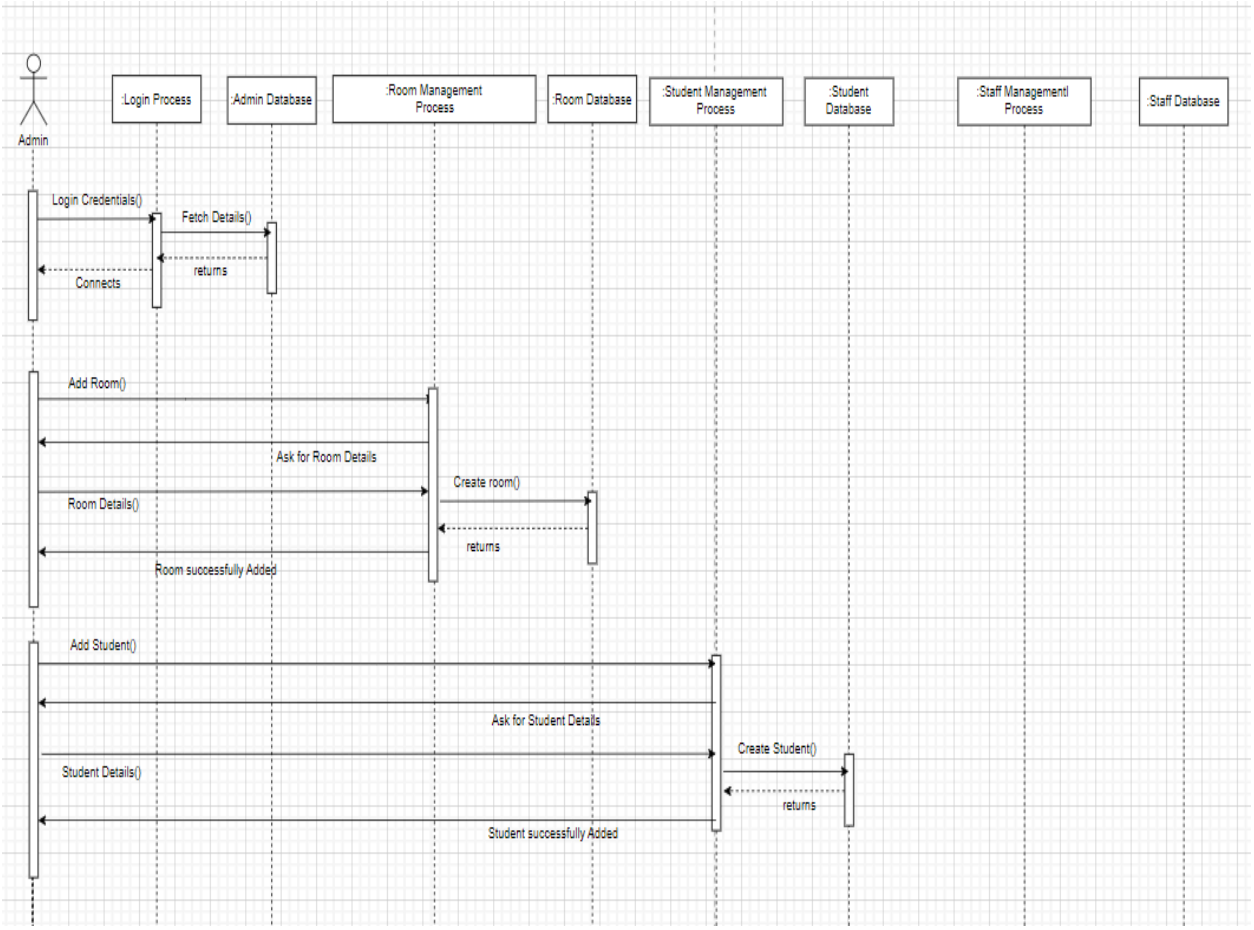
Staff Member :

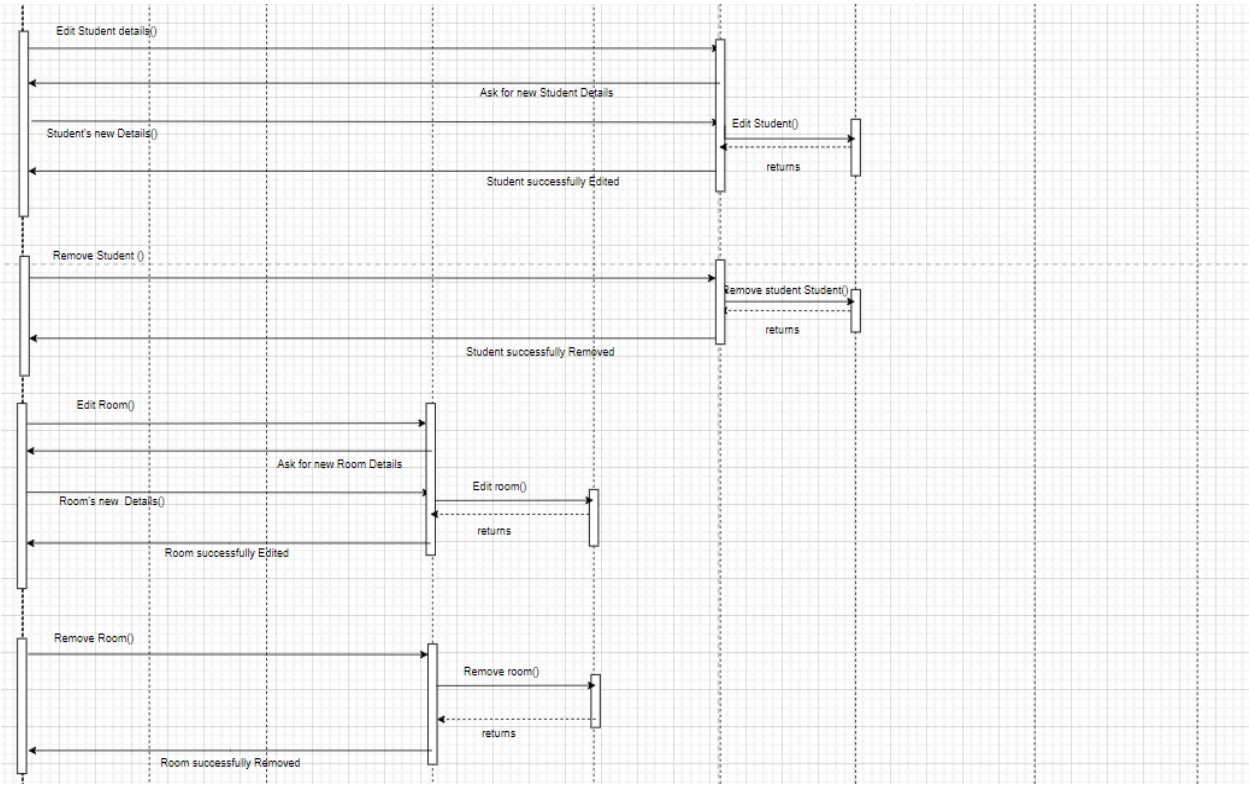

Staff
:Login Process
:Staff Database
:Complain Management Process
:Complain Database
:Room status Management Process
:Room Database
Login Credentials()
Fetch Details()
returns
Connects()
Display()
Review Complain()
Fetch Details()
Show complain details
returns
Process Complain()
Edit Status()
Complain status to Process
returns
Change room status()
Change Status()
returns
Successful change of room status
Logout()
Successful Logout

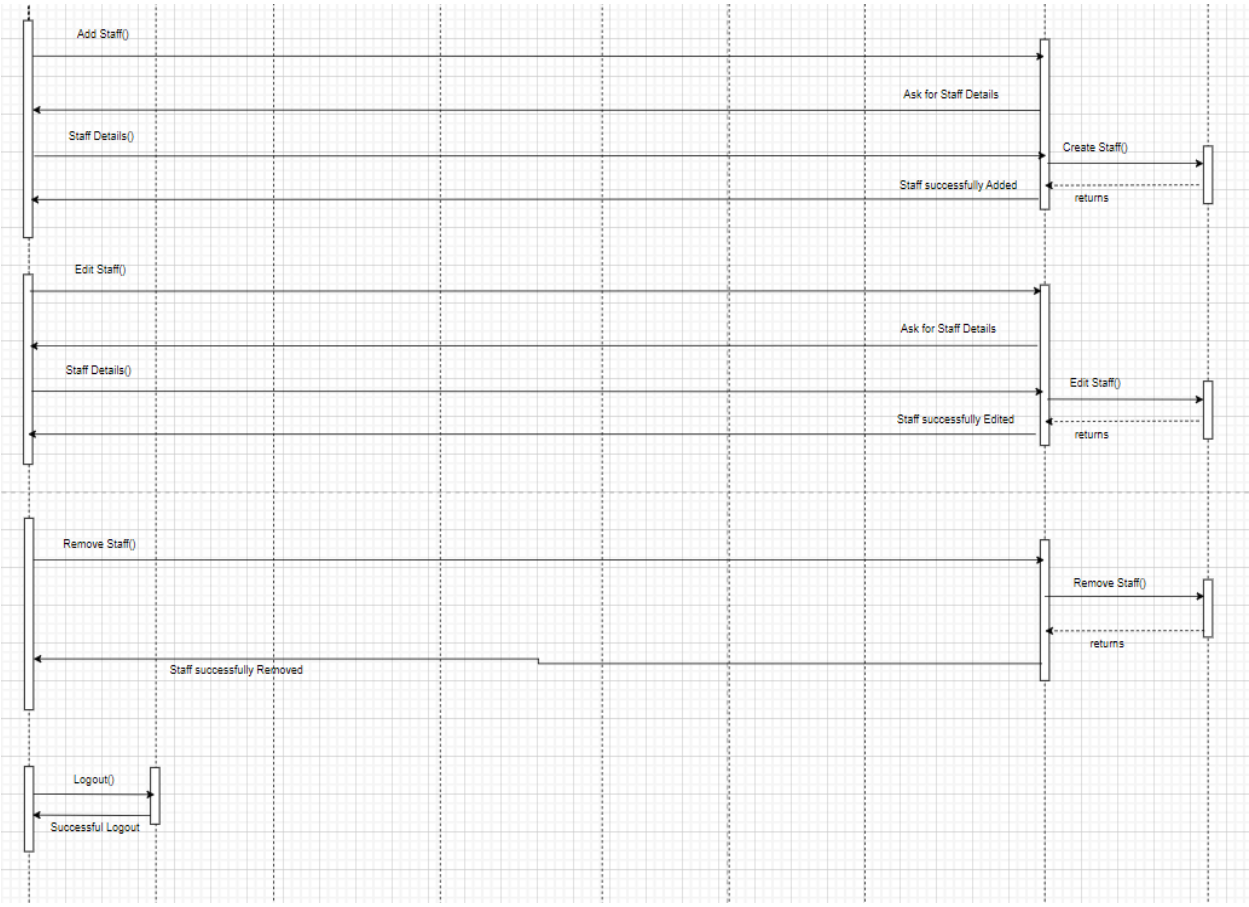
Student:


Student
:Login Process
:Student Database
:Complain Management Process
:Complain Database
Login Credentials()
Fetch Details()
returns
Connects()
Display()
Add Complain()
Ask for complain details
Send Details()
Add complain()
Complain Registered
returns
Change Complain status()
Change status()
returns
Complain status changed to resolve
Logout()
Successful Logout

Admin:


Admin
:Login Process
:Admin Database
:Room Management Process
:Room Database
:Student Management Process
:Student Database
:Staff Managementl Process
:Staff Database
Login Credentials()
Fetch Details()
returns
Connects
Add Room()
Ask for Room Details
Create room()
Room Details()
returns
Room successfully Added
Add Student()
Ask for Student Details
Create Student()
Student Details()
returns
Student successfully Added


Edit Student details()
Ask for new Student Details
Student's new Details()
Edit Student()
returns
Student successfully Edited
Remove Student ()
Remove student Student()
returns
Student successfully Removed
Edit Room()
Ask for new Room Details
Room's new Details()
Edit room()
returns
Room successfully Edited
Remove Room()
Remove room()
returns
Room successfully Removed


Add Staff()
Ask for Staff Details
Staff Details()
Create Staff()
Staff successfully Added
returns
Edit Staff()
Ask for Staff Details
Staff Details()
Edit Staff()
Staff successfully Edited
returns
Remove Staff()
Remove Staff()
returns
Staff successfully Removed
Logout()
Successful Logout